EL5 Worked Examples: Properties of Logarithms

Recording companion for the EL5 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What is happening inside the logarithm: multiplying, dividing, or a power? And what does each one turn into outside? Multiplying inside becomes adding outside. Dividing inside becomes subtracting outside. A power inside becomes a multiplier outside, and a negative exponent brings its sign with it. Condensing runs the same rules backward: coefficients move up into exponents first.

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Example 1: Rewrite 1

$$\log_2(xy)$$

Question

Which expression is equal to this logarithm?

- **A.** $\log_2 x \cdot \log_2 y$
- **B.** $\log_2 x - \log_2 y$
- **C.** $\log_2 x + \log_2 y$
- **D.** $\log_2(x+y)$

Answer and reasoning

Inside the logarithm, x and y are multiplied. Multiplying inside turns into adding two logarithms outside: $\log_2(xy) = \log_2 x + \log_2 y$. A quick number check: $\log_2(2 \cdot 8) = \log_2 16 = 4$, and $\log_2 2 + \log_2 8 = 1 + 3 = 4$. **The correct rewrite is $\log_2 x + \log_2 y$.**

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Example 2: Rewrite 2

$$\log_3(x/y)$$

Question

Which expression is equal to this logarithm?

- **A.** $(\log_3 x)/(\log_3 y)$
- **B.** $\log_3 y - \log_3 x$
- **C.** $\log_3 x + \log_3 y$
- **D.** $\log_3 x - \log_3 y$

Answer and reasoning

Inside the logarithm, x is divided by y. Dividing inside turns into subtracting outside, with the top first: $\log_3(x/y) = \log_3 x - \log_3 y$. Check with numbers: $\log_3(81/3) = \log_3 27 = 3$, and $\log_3 81 - \log_3 3 = 4 - 1 = 3$. Subtracting in the other order, $\log_3 y - \log_3 x$, gives -3. **The correct rewrite is $\log_3 x - \log_3 y$.**

Example 3: Rewrite 3

$$\log_5(x^4)$$

Question

Which expression is equal to this logarithm?

- **A.** $4\log_5 x$
- **B.** $(\log_5 x)^4$
- **C.** $\log_5 x+4$
- **D.** $\log_5(4x)$

Answer and reasoning

Inside the logarithm, x is raised to a power. A power inside comes out front as a multiplier: $\log_5(x^4)=4\log_5 x$. Check with $x=5$: $\log_5(5^4)=4$, and $4\log_5 5=4\cdot 1=4$. The other choices give 1, 5, and about 1.86. **The correct rewrite is $4\log_5 x$.**

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Example 4: Rewrite 4

$$\log_2(x+y)$$

Question

Which statement is correct about rewriting this logarithm?

A. $\log_2(x+y)=\log_2 x+\log_2 y$

B. $\log_2(x+y)=\log_2 x\cdot\log_2 y$

C. $\log_2(x+y)=(\log_2 x)/(\log_2 y)$

D. None of these. The properties only handle multiplying, dividing, or a power inside the logarithm.

Answer and reasoning

The properties of logarithms cover three things inside the logarithm: a product, a quotient, and a power. A sum inside has no rule. Test with $x=1$ and $y=3$: $\log_2(1+3)=\log_2 4=2$. Splitting it into a sum of logarithms gives about 1.58. Multiplying or dividing the two logarithms gives 0, because $\log_2 1=0$. None of them equals 2. **A logarithm of a sum cannot be split.**

Example 5: Expand 1

$$\log_3(x^2y^5)$$

Question

Which expression is the expanded form of this logarithm?

- **A.** $7\log_3(xy)$
- **B.** $2\log_3 x+5\log_3 y$
- **C.** $5\log_3 x+2\log_3 y$
- **D.** $\log_3(2x)+\log_3(5y)$

Answer and reasoning

First split the product: $\log_3(x^2y^5)=\log_3(x^2)+\log_3(y^5)$. Then bring each exponent out front as a multiplier. The 2 stays with x and the 5 stays with y. Adding the exponents to get 7 is a slip, because exponents of different letters never combine. **The expanded form is $2\log_3 x+5\log_3 y$.**

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Example 6: Expand 2

$$\log_5(a^3b^{-4})$$

Question

Which expression is the expanded form of this logarithm?

- **A.** $3\log_5 a - 4\log_5 b$
- **B.** $3\log_5 a + 4\log_5 b$
- **C.** $-3\log_5 a + 4\log_5 b$
- **D.** $(3\log_5 a)/(4\log_5 b)$

Answer and reasoning

Split the product: $\log_5(a^3)+\log_5(b^{-4})$. The power rule works with negative exponents too, and the exponent keeps its sign when it comes out front: $3\log_5 a+(-4)\log_5 b$, which is the same as subtracting $4\log_5 b$. It also makes sense because b^{-4} means $1/b^4$, and dividing inside is subtracting outside. **The expanded form is $3\log_5 a - 4\log_5 b$.**

Example 7: Expand 3

$$\log_4(x^5y^{-2}z^3)$$

Question

Which expression is the expanded form of this logarithm?

- **A.** $5\log_4 x+2\log_4 y+3\log_4 z$
- **B.** $5\log_4 x-2\log_4 y-3\log_4 z$
- **C.** $5\log_4 x-2\log_4 y+3\log_4 z$
- **D.** $-5\log_4 x+2\log_4 y-3\log_4 z$

Answer and reasoning

There are three factors, so there are three logarithms. Each exponent comes out front with its own sign: 5 with x, -2 with y, and 3 with z. Only the y term is negative, because only y has a negative exponent. The answer that starts with $-5\log_4 x$ has every sign reversed. **The expanded form is $5\log_4 x-2\log_4 y+3\log_4 z$.**

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Example 8: Expand 4

$$\log_6((a^3b^2)/c^4)$$

Question

Which expression is the expanded form of this logarithm?

- **A.** $3\log_6 a+2\log_6 b+4\log_6 c$
- **B.** $3\log_6 a-2\log_6 b-4\log_6 c$
- **C.** $(3\log_6 a+2\log_6 b)/(4\log_6 c)$
- **D.** $3\log_6 a+2\log_6 b-4\log_6 c$

Answer and reasoning

Everything on top of the fraction is multiplied, so those terms are added outside. Everything on the bottom is divided, so those terms are subtracted outside. Each exponent comes out front: $3\log_6 a+2\log_6 b-4\log_6 c$. Writing it as a fraction of logarithms treats the whole logarithm as a fraction, but dividing inside means subtracting outside. **The expanded form is $3\log_6 a+2\log_6 b-4\log_6 c$.**

Example 9: Condense 1

$$2\log_3 x+5\log_3 y$$

Question

Which single logarithm is equal to this expression?

- **A.** $\log_3(x^2+y^5)$
- **B.** $\log_3(10xy)$
- **C.** $\log_3(x^2y^5)$
- **D.** $\log_3(x^5y^2)$

Answer and reasoning

First move each coefficient up to be an exponent on its own letter: $2\log_3 x=\log_3(x^2)$ and $5\log_3 y=\log_3(y^5)$. Then adding two logarithms means multiplying what is inside: $\log_3(x^2)+\log_3(y^5)=\log_3(x^2y^5)$. Adding inside, or multiplying the coefficients, does not follow any rule. **The single logarithm is $\log_3(x^2y^5)$.**

Example 10: Condense 2

$$4\log_2 x - 3\log_2 y$$

Question

Which single logarithm is equal to this expression?

- **A.** $\log_2(x^4/y^3)$
- **B.** $\log_2(x^3/y^4)$
- **C.** $\log_2(x^4 - y^3)$
- **D.** $\log_2((4x)/(3y))$

Answer and reasoning

Move the coefficients up first: $\log_2(x^4) - \log_2(y^3)$. Subtracting two logarithms means dividing what is inside, with the first one on top. CheckIt may write this answer as $\log_2(x^4y^{-3})$. That is the same thing, because $y^{-3} = 1/y^3$. **The single logarithm is $\log_2(x^4/y^3)$.**

Example 11: Condense 3

$$3\log_5 a - 2\log_5 b + 4\log_5 c$$

Question

Which single logarithm is equal to this expression?

- **A.** $\log_5((a^3b^2)/c^4)$
- **B.** $\log_5((a^3c^4)/b^2)$
- **C.** $\log_5((a^2c^3)/b^4)$
- **D.** $\log_5(a^3/(b^2c^4))$

Answer and reasoning

Move each coefficient up: $\log_5(a^3) - \log_5(b^2) + \log_5(c^4)$. Added terms are multiplied on top, and the subtracted term goes on the bottom. So a^3 and c^4 are on top, and b^2 is on the bottom. Only the term with the minus sign goes into the denominator. **The single logarithm is $\log_5((a^3c^4)/b^2)$.**

Example 12: Condense 4

$$\log_2 x + 3\log_2 y - 5\log_2 z$$

Question

Which single logarithm is equal to this expression?

- **A.** $\log_2(xy^3z^5)$
- **B.** $\log_2(y^3/(xz^5))$
- **C.** $\log_2((x^3y)/z^5)$
- **D.** $\log_2((xy^3)/z^5)$

Answer and reasoning

The first term has no visible coefficient, which means the coefficient is 1, so x stays x^1. Move the other coefficients up: $\log_2 x + \log_2(y^3) - \log_2(z^5)$. The two added terms multiply on top, and the subtracted term goes on the bottom. **The single logarithm is $\log_2((xy^3)/z^5)$.**

Example 13: Slip 1

$$\log_3(x^5)=(\log_3 x)^5$$

Question

A student wrote this line. What is the mistake, if any?

A. The base should change from 3 to 5.
B. The 5 should multiply the log, not be an exponent on the log.
C. A power inside a logarithm should turn into addition outside.
D. There is no mistake.

Answer and reasoning

A power inside a logarithm comes out front as a multiplier: $\log_3(x^5)=5\log_3 x$. The student raised the logarithm itself to the 5th power, which is a different number. Check with $x=3$: $\log_3(3^5)=5$, but $(\log_3 3)^5=1^5=1$. **The correct line is $\log_3(x^5)=5\log_3 x$.**

Example 14: Slip 2

$$\log_2(x/y)=(\log_2 x)/(\log_2 y)$$

Question

A student wrote this line. What is the mistake, if any?

A. The base 2 should have been dropped from both logs.
B. The two logarithms should have been added, not divided.
C. Dividing inside means subtracting logs, not dividing.
D. There is no mistake.

Answer and reasoning

Division inside becomes subtraction outside: $\log_2(x/y)=\log_2 x-\log_2 y$. Check with $x=8$ and $y=2$: $\log_2 4=2$ and $3-1=2$, but $3/1=3$. **The correct line is $\log_2(x/y)=\log_2 x-\log_2 y$.**

Example 15: Slip 3

$$\log_2(8x)=3+\log_2 x$$

Question

A student wrote this line. What is the mistake, if any?

A. The 8 should come out front as a multiplier of the log.
B. The right side should be 8 times log base 2 of x.
C. There is no mistake.
D. A log of a product cannot be split into two logs.

Answer and reasoning

Multiplying inside becomes adding outside: $\log_2(8x)=\log_2 8+\log_2 x$. And $\log_2 8=3$, because $2^3=8$. Check with $x=2$: $\log_2 16=4$, and $3+\log_2 2=3+1=4$. **There is no mistake.**

Example 16: Slip 4

$$\log_5(a^{-2}) = 2\log_5 a$$

Question

A student wrote this line. What is the mistake, if any?

A. There is no mistake.
B. The 2 should have been added to the logarithm instead.
C. The logarithm should have been divided by 2.
D. The negative sign was lost: it should be negative 2.

Answer and reasoning

The exponent keeps its sign when it comes out front: $\log_5(a^{-2}) = -2\log_5 a$. Check with $a = 5$: $\log_5(5^{-2}) = -2$, but $2\log_5 5 = 2$. **The correct line is $\log_5(a^{-2}) = -2\log_5 a$.**

Example 17: Slip 5

$$3\log_2 x - \log_2 y = \log_2((3x)/y)$$

Question

A student wrote this line. What is the mistake, if any?

A. Move the 3 up as an exponent on x before combining.
B. The minus sign should have turned into a plus sign here.
C. The fraction should have been flipped upside down.
D. There is no mistake.

Answer and reasoning

Move the coefficient up first: $3\log_2 x = \log_2(x^3)$. Then subtracting becomes dividing: $\log_2(x^3) - \log_2 y = \log_2(x^3/y)$. Check with $x=2$ and $y=1$: the left side is 3, the correct right side is $\log_2 8 = 3$, and the student's right side is $\log_2 6 \approx 2.58$. **The correct line is $3\log_2 x - \log_2 y = \log_2(x^3/y)$.**